

AWI–ESM3 v3.4 PI-control tuning campaign — summary

What to adopt, what the lever hunt has exhausted, and the three structural problems underneath

OpenIFS 48r1 / FESOM2 / LPJ-GUESS, TCO95L91

58 atmosphere-only lever runs, 30 coupled runs

2026–08–11

Abstract

The short reading of a campaign whose full record runs to 110 pages (`report.tex`). It keeps what still constrains a decision and drops what died on the way. Three objectives: the **global radiative imbalance**, the **Southern Ocean**, and the **Siberian cold bias / boreal forest**.

Adopt S4. RCL_INPPMIN 70000 $\rightarrow$ 50000, one namelist number on the existing stack, is the best configuration the campaign has produced: it takes the Southern Ocean surface shortwave bias from +3.66 to -0.96 W m^{-2} and global net TOA from +0.64 to **+0.25** — nearest the pre-industrial target of anything run — at *no* cost in either Siberian season or the tropics. It finished on 2026–08–09 and sat unevaluated for two days while three more expensive arms were built.

The lever hunt beyond it is exhausted. Rounds 26–28 built and ran a marine-biogenic-CCN scheme, a temperature-selective cirrus lever, and a cloud-overlap arm. All three work, none is adoptable: scored globally rather than in the Southern Ocean box, they move the tropics as much as the Southern Ocean and are **global dimming knobs wearing a regional label**. The one genuinely new physical result — RSNOWLIN2, which closes the entire tropical longwave deficit, five times any previous lever — *warms* the Southern Ocean and worsens the global budget.

Siberia’s two halves have separated. Coupled 11E is the first arm to warm Siberian summer without a winter penalty and is essentially the atmospheric answer. The *forest* is untouched: the coupled model inherits a real boreal forest and destroys 92 % of the larch in fifty years. But offline 2000-year spin-ups now calibrate the fix — +2 K of warming recovers 59 % of the forest gap, monotonically and without saturating — so closing the $\approx 2 \text{ K}$ cold bias should recover most of it. The campaign’s founding premise, measured rather than assumed.

Three structural problems surfaced that no lever can address and that change how results should be read: the AMIP runs are **not at 1850 forcing** (Krakatoa is visible in them); the Southern Ocean’s “band mean” and its “RMSE” are **different fields** that disagree about which arm is best; and the coupled amplification is $1.40\times$ **via sea-ice albedo, not cloud**, so an AMIP cloud number is not what survives coupling. §17 is what to do next.

1 Scorecard

objective	now	target	status
global net TOA (AMIP, S4)	+0.25	0	nearly closed ; ± 0.15 of it is a forcing artefact
global net TOA (coupled, 11E)	+0.79	0	open; 11G tests S4 coupled
SO surface SW bias (S4)	-0.96	0	nearly closed from +3.66
SO TOA SW CRE bias (S4)	+3.02	0	half closed from +7.69; <i>disagrees with the surface</i>
Siberia JJA T2m	-0.9 K	0	half closed; land levers exhausted
boreal forest, coupled	TREE 0.083	≈ 0.42	not solved ; needs the cold bias closed

Carried forward: G4 + K1 + P3/SWEMIN= 15 + D2b + **S4**. All namelist-only. **Not** carried: RSNOWLIN2, DMS, RCL_OVERLAPLIQICE, and the whole round-26/27 lever line.

2 The decision: S4, and why nothing beyond it survives

Every 44-year arm scored on all three constraints at once — Southern Ocean *surface* bias against CERES over ocean, the tropics, both Siberian seasons — with global net TOA carried alongside so that a global dimming knob cannot pass as a regional fix:

	SO sfc	SO rmse	Δ tropics	Δ JJA	Δ DJF	global TOA
control	+3.66	7.04	—	—	—	+0.64
S4 inppmin50k	−0.96	5.31	+0.26	+1.03	+0.05	+0.25
LX1 +RSNOWLIN2	−0.52	5.15	+0.75	+1.29	+0.54	+1.08
P5 (adopted stack)	+0.74	5.12	+0.35	+1.15	−0.06	+0.66
LX3 +DMS+INP	−4.75	7.31	−0.97	+1.33	+0.26	−0.88
LY2 +ov10.10	−4.75	6.81	−0.32	+0.61	+2.14	−0.58
A1a ov1= 0.10	−2.32	5.70	−1.06	−0.75	+0.83	−1.25

LX1 is nominally closer to zero on the Southern Ocean and pays 0.83 W m^{-2} of global imbalance for the last 0.4 — a bad trade. LX3 and LY2 reach the Southern Ocean by dimming the planet. A1a damages the tropics *and* Siberian summer, confirming its round-3 rejection on both counts. **S4 is the only arm that improves the Southern Ocean while landing global net TOA near target and costing nothing elsewhere.**

2.1 Why the new levers are global knobs

Mapped globally rather than in the Southern Ocean box, LX3 moves the **tropics** -4.05 in TOA SW CRE against the Southern Ocean’s -5.61 — the same order, and near-uniformly across every basin. It survives on *net* only because RSNOWLIN2 adds $+2.46$ of global longwave cloud forcing in compensation: the configuration makes the planet dimmer and more greenhouse-y and calls the residual a Southern Ocean fix. DMS’s seasonal selectivity at one year — genuinely striking, -7.25 in DJF where the deficit is worst — **does not become spatial selectivity at 44 years.**

3 The global radiative imbalance

3.1 The target is zero, and the reason is snow enthalpy

The target was inherited as -0.16 W m^{-2} , which was never derived — it is whatever an HR piControl happened to settle at. A target away from zero is only justified if the atmosphere fails to conserve energy, because equilibrium requires $\text{SFC} = \text{TOA} + S$ for a spurious atmospheric source S , so zero *ocean* drift would need $\text{TOA} = -S$. Measured on 10A over its last decade, using the release tool’s own formula: net TOA $+0.854$, net surface $+0.877$, **residual** -0.023 W m^{-2} . The atmosphere conserves to 0.02, so “ocean not drifting” and “system not drifting” are the same condition and both put the target at 0.

Snow enthalpy does nearly all of that reconciliation: $\text{sf} \times 3.3355 \times 10^8$ is 0.966 W m^{-2} , and omitting it makes the surface look like $+1.84$ against a TOA of $+0.85$ — the artefact that motivated -0.16 -style fudges in the first place.

3.2 Where it lives, and what is left

Period-clean, the imbalance is not global but regional: the Southern Ocean band contributes $+0.74$ of it on its own, through a $+7.46 \text{ W m}^{-2}$ band error. The tropics are *negative* — -0.67 against CERES — so they need more absorbed energy, not less. That asymmetry is why a global brightness knob cannot fix the budget: it moves both bands the same way and the two need opposite corrections.

S4 takes the AMIP imbalance to $+0.25$. Of the remaining 0.25, roughly 0.15 is the transient-forcing artefact of §13, so the *model’s* residual imbalance may be nearer $+0.1$ than $+0.25$. That cannot be confirmed without the fixed-forcing control run.

4 The tropical longwave: found, and unusable

The four-way attribution splits the two remaining regional failures by channel: the Southern Ocean fails through **shortwave** cloud ($+7.45$), the tropics through **longwave** cloud (-2.16). For most of the campaign nothing moved the second at all — scored across all 50 archived arms, the largest positive response was A1a’s $+0.373$, 17% of the deficit, and only as a by-product of a lever that wrecks everything else. The term is extraordinarily stiff: interannual sd 0.101 W m^{-2} , so the deficit is $51\times$ its own detection threshold.

4.1 RSNOWLIN2: the largest tropical longwave response ever measured here

RSNOWLIN2 is the *temperature exponent* of ice→snow autoconversion, $ZZCO = PTSPHY \cdot RSNOWLIN1 \cdot e^{RSNOWLIN2(T-T_0)}$. Raising it $0.030 \rightarrow 0.04$ slows the conversion 31 % at a 200 K cirrus temperature but only 11 % at 250 K, so it builds **cold thin cirrus rather than thickening anvils** — and thin cirrus is longwave-dominant while a thick anvil is bright in both. The contrast is measured: DETRPEN gives an LW:SW ratio of 0.44, RSNOWLIN2 1.17.

Result: $+2.74 \text{ W m}^{-2}$ of tropical LW CRE, **127 % of the deficit**, five times anything else in the archive, with tropical and global net TOA both below threshold at one year.

4.2 Why it cannot be adopted

At 44 years it adds +0.61 to Southern Ocean *net* forcing — its +2.6 of SO longwave swamps its shortwave — and +0.42 to global net TOA. On the bare model it was neutral; on the adopted stack the sign flips. **The campaign’s two remaining regional targets are in direct tension through this one parameter:** it closes the tropical longwave deficit partly by putting cirrus over the Southern Ocean, where the model already traps too much.

The value 0.04 is not invented — it is the setting in the v3.4.2 TCO95L91-DARS2 runscripts, part of a jointly-tuned eight-parameter cloud stack for this exact resolution that no `fort.4` on disk has ever executed. That stack was found only because `preflight.py` was scanning one runtime tree and one runscript directory; it now scans every one, and reports values rather than mentions.

5 The Southern Ocean

5.1 Two fields, and they disagree

The campaign quotes the Southern Ocean band mean from **TOA cloud radiative effect** and its “SW RMSE” from **surface net shortwave over ocean**. These are different quantities and they do not agree about which arm is best.¹

	TOA SW CRE bias	surface net SW bias
control	+7.69	+3.66
P5	+4.98 (<i>bad</i>)	+0.74 (<i>nearly right</i>)
LX3	−0.63 (<i>nearly right</i>)	−4.75 (<i>bad</i>)

Closing a TOA gap of $\approx 5 \text{ W m}^{-2}$ costs the surface ≈ 5 , and it did, almost exactly. **The two constraints cannot both be satisfied by adding cloud reflection.** That is an *atmospheric shortwave absorption* signature — the model would need cloud that reflects more at TOA without dimming the surface proportionally, which is absorbing cloud, not more cloud. The surface error also exceeds the TOA error *globally* (14.4 against 11.5), so this is not a Southern Ocean artefact. Until it is resolved, the campaign has been optimising against whichever metric happened to be quoted.

¹LX3 = P5 + RSNOWLIN2 = 0.04 + DMS + RCL_INPPMIN = 50 000. Screened at 44 years and *not* adopted — it reaches the Southern Ocean by dimming the planet (§2). It is quoted here and in Fig. 1 only because it is the arm that separates the two fields most cleanly.

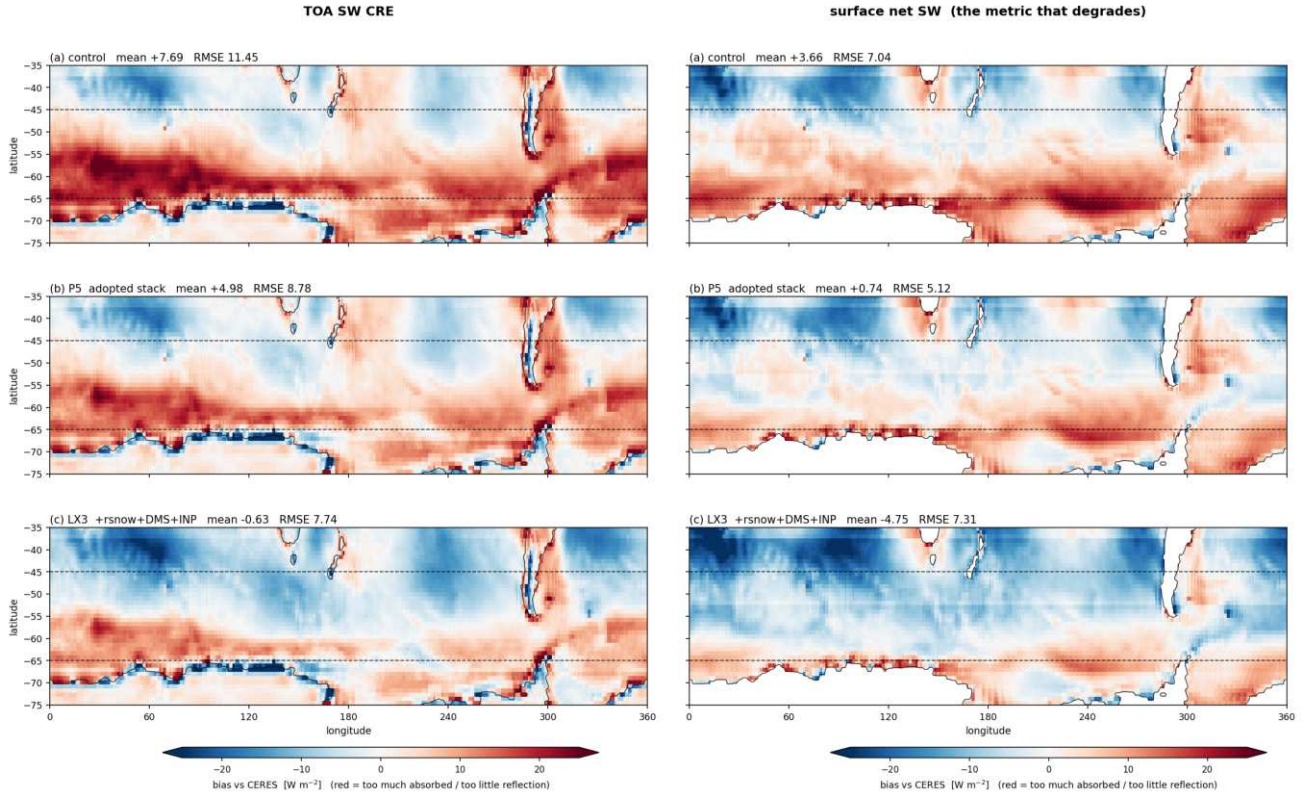


Figure 1: The Southern Ocean, and the two fields that disagree. Bias against CERES for the control, the adopted stack P5, and the rejected LX3; TOA cloud forcing left, surface net shortwave right, one shared scale down each column, dashed lines the scored 45–65S band. P5 (middle) takes the surface nearly to zero while leaving TOA 5 W m^{-2} short; the bottom row shows what closing that TOA gap costs — a *uniform* -4.75 at the surface. The residual structure — the red band along 55–65S and the Antarctic coastal strip — is present at similar strength in all three rows: the levers act as a near-uniform multiplier, not a spatial correction.

5.2 What actually moves it, on NET

Longwave cancels a very consistent 25–29 % of the shortwave gain across the whole 44-year archive, so $\text{SO SW} \times 0.72$ is a reliable rule for ordinary cloud levers. Best net levers: A1a -4.57 , D2a -3.60 , S4 -3.47 . The round-26/27 levers break the rule — **RSNOWLIN2** adds $+2.6$ of Southern Ocean *longwave*, enough to flip its net contribution positive on the adopted stack.

5.3 The cloud-amount deficit is still open

Two thirds of the band error is a ≈ 6 pp cloud-*amount* deficit. S4 adds $+0.105$ pp. Nothing in 58 runs has closed a meaningful fraction of it, and no identified lever reaches it.

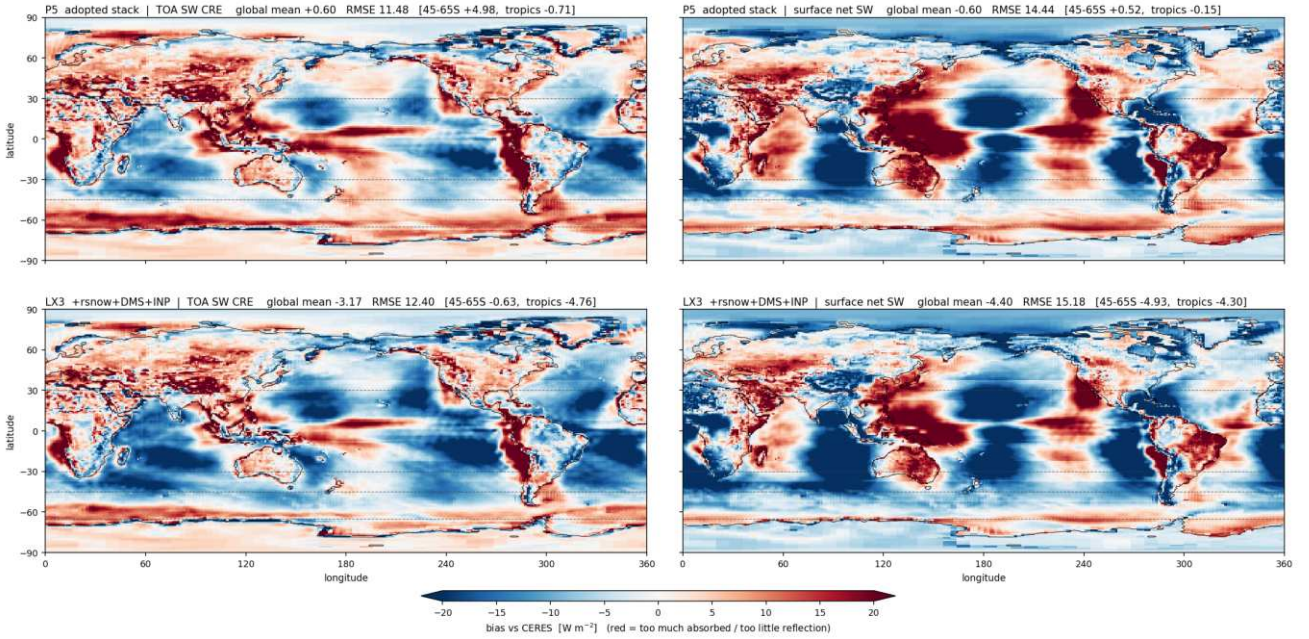


Figure 2: Why LX3 is not a Southern Ocean lever. Shortwave bias against CERES for P5 (top) and LX3 (bottom); TOA cloud forcing left, surface net shortwave right. The change between rows is near-uniform blue in *every* basin: the tropics move -4.05 against the Southern Ocean’s -5.61 . Global RMSE barely moves (TOA $11.48 \rightarrow 12.40$) because the error field is set by the tropical Pacific dipole, the stratocumulus decks and the land signals — none of which any lever touches. Note also that the surface field (right) carries a *larger* error than TOA (left) everywhere, an atmospheric-absorption signature.

6 The land surface: how the boreal half was actually solved

Twelve of the campaign’s twenty-eight rounds were spent here, and this is where the largest temperature gains of the campaign came from. The arc is worth stating compactly because the adopted stack is almost entirely land.

6.1 The bias is LPJ-GUESS, and it acts by two routes

A purpose-built no-LPJG reference — same CORE3 ocean, same TCO95 atmosphere, HTESSEL with prescribed satellite vegetation, 30 coupled years — differs from the baseline in **LPJG alone**. Siberian high-vegetation cover is 0.63 without and **0.02** with; the LPJG run is -3.1 K colder in DJF. Removing LPJG also *improves* global 2m-T RMSD (1.514 vs 1.717) and CMPI (0.738 vs 0.745) at a TOA imbalance only 0.2 W m^{-2} higher, so the warming is nearly free of the radiation objective.

The seasonal fingerprint identifies the mechanisms. The cold effect is present in every month except high summer and peaks in the *dark*: November -3.9 K, April -3.3 K, July–August ≈ 0 . A pure snow-albedo mechanism cannot do November. The second route is **roughness**: needleleaf forest has $z_{0m} = 2.0$ m against 0.034–0.10 m for the tundra and grass that replace it, and a smoother surface decouples from the atmosphere in the stable boundary layer. The data carry exactly that signature — a $+2.1$ K stronger $T_{925} - T_{2m}$ inversion, $+0.7$ K stronger surface decoupling, -4.7 W m^{-2} of down-welling longwave, and **unchanged cloud** ($\Delta \approx 0.00$, the negative control). Spatially $r(\Delta cvh, \Delta inv) = -0.60$ against $r(\Delta cvh, \Delta cloud) = -0.05$.

A residual survives with the forest intact: the model’s winter inversion is $+1.4$ K too strong against ERA5, *entirely surface-trapped* (the 925–1000 hPa bias is $+0.07$), and it explains $r = +0.76$ of where the model is too cold. Circulation ($r = 0.27$) and cloud fraction ($r = 0.00$) are excluded. That is the IFS-family under-mixing stable boundary layer, and its lever is orthogonal to everything tried — **it remains unattacked**.

6.2 Half the summer shortwave deficit is albedo, and it is melt timing

Decomposed against CERES on the identical box and mask, the 12.5 W m^{-2} Siberian JJA surface-SW deficit is 7.0 cloud and **5.5 surface albedo** — and the albedo half is almost entirely June (13.4 of it, against 1.7 in July). Temperate and tropical land is correct to ± 0.004 , so the albedo scheme is sound; the error is high-latitude ($+0.094$ on Siberian tundra) with **Fennoscandia, the same needleleaf type, correct at $+0.002$** .

It is not a brightness error. Monthly, the model is too *dark* in April (-0.06), perfect when snow-free in July ($+0.004$), too bright in June ($+0.09$) and September ($+0.09$): **the snow season is too long at both ends**. Against ERA5 the accumulation is right to 7–9% with the monthly increments tracking almost exactly, and the whole divergence is spring — **the snowpack peaks a month late** (April still gaining $+9.4$ mm while ERA5 has begun melting) and **May melts 30% too slowly** (33.5 against 47.5 mm). Satellite confirms it independently: melt-out 13 days late.

That closes a runaway. In May the *downwelling* flux is nearly right (-3.5) while the net is -13.9 : pure albedo. Late melt $\rightarrow$ more May snow $\rightarrow$ higher albedo $\rightarrow$ less absorbed $\rightarrow$ less melt energy $\rightarrow$ snow survives into June. The model melts slower *despite* having darker snow than ERA5 — the brightness term works against the bias and the extent term wins.

Two cause-and-effect checks kept this honest. February–April is simultaneously too cold *and* too dark, so the winter cold is **not** albedo-driven. And insolation-weighted over global land, the albedo-induced SW loss correlates with the T2m bias at only $r = +0.145$ — cells where the model is too *dark* are still -1.16 K cold. **Albedo is worth fixing and is not the root cause**.

6.3 The lever ladder, and what it cost

	what it changes	Siberian JJA	outcome
F4	RVRSMIN(3,4) 250 $\rightarrow$ 1000	+0.749	adopted ; best single lever
F5	F4 + three more	+0.746	$\equiv$ F4; one pathway, saturated
F1	RVZOH = $z_{0m}/10$	+0.185	marginal — <i>predicted strongest</i>
G4	RVRSMIN pushed further	+0.952	adopted , then capped
I3	tundra albedo	+1.322	rejected: DJF -2.44 , Arctic TOA $+1.7$
K1	RVVEGALB crops/semidesert + soil	+1.036	adopted ; best global RMSE 1.524
H	snow-cover fraction, first attempt	—	rejected: prediction falsified
LAMSK	snow-tile skin conductivity	—	falsified; the falsifier was on record
L	tundra “moss proxy”	—	falsified, monotonically the wrong way
P3/P5	ECE_SNOW_SCF = 3, SWEMIN = 15	+1.34	adopted

Three things in that table matter more than the numbers. **F4 was anchored on an observation, not a sweep** — the modelled Siberian JJA Bowen ratio is 0.43 against 0.5–1.5 observed — and it moved Bowen to 0.60, into the observed range without overshooting, while costing the tropics nothing (42.68 against 42.61). **The strongest lever was disqualified by a season nobody was scoring**: B5, the previous best, carried a hidden DJF -0.72 K, invisible for the whole campaign because only JJA was evaluated. DJF has been a standing guardrail since. And **K1’s negative result was worth more than its $+0.089$ K**: it removed 74% of a directly measured global land albedo bias and bought a sixth of the target, which *closes out land albedo* as the explanation for the universal cold bias — at a measured sensitivity of 0.116 K per W m^{-2} against 0.12 assumed.

6.4 The snow scheme, rebuilt from observations

Route A — transpiration to cloud — is closed: the vegetation levers act through surface exchange, not cloud. Route B — snow to albedo — is where the gains came from, and it required replacing the scheme outright. The inherited **tanh** depletion has range (0, 1) *open at 1*, so complete snow cover is not representable at any parameter setting, while RIHMI-WDC snow courses report exactly 10/10 in 99.74% of 14369 Siberian DJF surveys. Against 174676 station observations the **tanh** scored -26.3 against an as-released -5.3 ; coupled, it drove the soil **22.7 K** cold and collapsed the soil-to-snow January gradient from 22.4 K to 1.8 K, which is physically impossible. ECE_SNOW_SCF = 3 replaces it with $\min(1, (d/d_c)^b)$, d_c split by vegetation type and fitted to 36492 snow-course surveys, with SWEMIN flooring d_c at a minimum snow *mass* — the right currency, since the failure it fixes is a heat-capacity collapse when a sub-millimetre dusting is given full cover.

It passed both pre-registered falsifiers and beats the scheme-off baseline on every soil metric (bias -1.13 against -1.52 , RMSE 3.12 against 3.28), and the winter penalty that had followed every snow lever since round 15 is gone. **It also costs the autumn**: the fitted d_c for fresh low-density snow is ~ 2 mm, correct at the 1–2 km scale the snow courses measure and wrong at grid scale, so September Rutgers cover goes $2.4\times$ worse ($+0.259$ against $+0.107$). The offline sweep puts the whole-season optimum at SWEMIN = 30, but coupled and in AMIP 15 wins — the offline score holds the albedo $\rightarrow$ melt $\rightarrow$ SWE feedback fixed, and the feedback is the point.

The single most useful thing the land work produced is an observational anchor. 105 RIHMI-WDC stations of Siberian soil temperature, QC-filtered, turned two rounds of plausible-looking snow tuning into a decided question in one figure.

6.5 The ceiling

Boreal screen temperature is remarkably insensitive to surface shortwave: $0.02\text{--}0.10\text{ K per W m}^{-2}$ across the radiative levers, and one manages the wrong sign. Closing the *entire* 14 W m^{-2} deficit would buy $\approx 1.3\text{ K}$ of a $\approx 2\text{ K}$ bias. The energy goes to sensible heat and the temperature hardly responds. **The cloud error was never the whole story:** the land levers stop at half the bias, and the residual sits in the global tropospheric cold bias that no land lever reaches.

7 Siberia and the boreal forest

7.1 The atmospheric half is done

Coupled **11E** (11D at SWEMIN= 15 plus K1) is the first coupled arm to warm Siberian summer *without* a winter penalty — JJA $+0.73$, DJF $+0.39$, DJF soil $+0.39$ against the baseline, where every previous gain came with a winter cost. Against 11D on a common 30-year window it gives DJF soil **$+0.90\text{ K}$** , resolved against a ± 0.37 threshold: **the AMIP choice of SWEMIN= 15 transfers to the coupled model**, as predicted from P5-versus-P6.

A caution that cost a wrong sign: at a 10-year window the 11E–11D DJF difference came out -0.54 ; at 30 years it is $+0.40$. Coupled comparisons need ≥ 30 years.

7.2 The forest is not

Siberian box, 473 cells	BNS (larch)	TREE	tree>0.5	grass>tree
offline CRUNCEP (the restart state)	0.227	0.415	50.3 %	29.2 %
11E at year 1350 (start)	0.213	0.394	47.8 %	28.3 %
11E after 50 coupled years	0.017	0.083	2.1 %	69.6 %

The coupled model inherits a real boreal forest and dismantles it in fifty years — 92 % of the larch — from a healthy state, while clearing the larch growing-degree gate on 91 % of cells. It is a transient toward a much poorer equilibrium, not an endpoint: forest composition takes centuries.

7.3 And warming fixes it — now calibrated

Three 2000-year offline LPJ-GUESS spin-ups at TCO95:

offline equilibrium	BNS	TREE	tree>0.5	BNS>0.1
CRUNCEP (observed forcing)	0.227	0.415	50.3 %	36.2 %
AMIP forcing	0.034	0.202	18.2 %	7.4 %
AMIP +1 K	0.058	0.261	24.5 %	12.5 %
AMIP +2 K	0.099	0.327	31.7 %	19.7 %

Monotonic, not saturating, and +2K recovers 59 % of the gap. With the AMIP cold bias at $\approx 2\text{ K}$ period-clean, **closing it should recover most of the forest**. Note grass>tree stays at 43–45 % across all three against CRUNCEP’s 29 %: warming raises tree cover *without* dissolving the competition. Forcing is the driver; grass-versus-larch remains the mechanism.

7.4 The land levers are exhausted

The adopted stack recovers 1.15 K of a $\approx 2\text{ K}$ period-clean bias and stops. The residual sits in the global tropospheric cold bias, which no land lever reaches. Three offline competition experiments (BNS_GREFF0035, IBS_GREFF012, C3G_PSTEMLOW12) are unread and belong to the final land round.

8 Southern Ocean and Siberia are not in competition

Across the coupled archive the two original faults appeared to trade off almost linearly: every W m^{-2} of TOA improvement cost $\approx 0.6\text{ K}$ of global cold. **That line is a property of one lever family, not of the model.** It was drawn by aggressive ocean mixing, which lowers TOA by cooling the planet — necessarily coupling the two objectives.

Levers acting on disjoint geography and disjoint physics escape it. D2b (Southern Ocean ice nucleation) and F4/G4 (boreal surface exchange) superpose additively to within noise on every metric, and S4 — the arm now adopted — improves the Southern Ocean surface with Siberian JJA +1.03 and DJF +0.05, i.e. no interaction at all.

The round-26/27 levers are the counter-example that proves the rule: they are *not* geographically disjoint, and they behave exactly as the ocean-mixing family did, trading one objective against another. **Selectivity is a property to be verified globally, not assumed from the mechanism** — DMS looked seasonally selective at one year and turned out to be a global knob at 44.

9 The shared limiter: the tropopause and the cold point

Both remaining objectives run into the same upper-atmosphere error, and it is not what it first looked like.

~~The extratropical cold and dry pair is a displaced tropopause.~~ The WMO lapse-rate tropopause sits on the *same level* as ERA5 in all five latitude bands. What is there instead is a *sharper transition*: the model cools +1.8 K km⁻¹ faster than ERA5 through 300–250 hPa at the poles and 1.5–1.9 slower above 200.

And the cold and dry halves are **one error, not two**. Stratospheric water vapour is set by the temperature of the tropical cold point, which freeze-dries air entering the stratosphere; the Brewer–Dobson circulation then carries it poleward. The model’s tropical cold point is 193.44 K against ERA5’s 195.51 — -2.07 K — and Clausius–Clapeyron over ice predicts -28.5 % of entry-air drying against a measured -46.9 % at 100 hPa. **A 2 K cold-point bias accounts for three fifths of a 47 % stratospheric dry bias.**

That is much better news than the alternative. Two independent cancelling errors would have to be fixed together or not at all, and fixing either alone moves net TOA several W m⁻² the wrong way. One error with one entry point means **a lever that warms the tropical cold point should move both**, in the right direction, at once. No such lever has been identified, and the remaining 39 % — transport, or UTLS microphysics — is untraced.

Caveat carried forward: ERA5’s upper-level humidity is weakly observationally constrained, so the -46.9 % denominator is the softest number here. The -2.07 K cold point rests only on temperature, which ERA5 constrains well.

10 The marine-CCN line: what was built, and why it is shelved

Two rounds went into this and it produced real physics, so it is documented rather than merely dismissed.

What the model was missing. The aerosol–cloud path was traced end to end. Genthon (1992) supplies sea-salt CCN from wind and MACv2-SP supplies anthropogenic plumes; there was **no marine biogenic CCN anywhere**. The consequence was measurable: Southern Ocean droplet number is right in the *mean* ($N_d \approx 73 \text{ cm}^{-3}$ against an observed 60–120) but its **seasonal phase is inverted** — lowest in DJF (68.5), highest in JJA (76.1), where observations peak in DJF with the bloom and Cape Grim sees a 2–3× summer/winter ratio.

What was built. DMS-Rev3 (Hulswar et al. 2022) was ingested through ocp-tool onto the reduced Gaussian grid as an eighth ICMCL field — zero holes at TCO95 — and wired into `radpar.F90` after SURFRAD fills PCCNO: Saltzman et al. (1993) Schmidt number, Nightingale et al. (2000) transfer velocity, $F = kC_w$, and PCCNO += $S \cdot F_S$ with S a namelist knob defaulting to zero. It flows through the Martin et al. (1994) quadratic to droplet number and effective radius — the Twomey effect. Everything is gated so that with the switch off the configuration reproduces **bit-for-bit**, which was verified.

What it showed. Woodhouse et al. (2010) measure dN/dF from -43 to +166 cm⁻³ per mg S m⁻² d⁻¹, too wide to pick from, so the range was bracketed. At the top of it:

$S = 166$, SW CRE	DJF	MAM	JJA	SON	annual
SO 65–45S	-7.25	+0.03	-0.63	-1.14	-2.25
tropics	-1.56	-2.17	-2.44	-2.39	-2.14

The DJF response is 67 % of the +10.86 W m⁻² austral-summer deficit, in the season the deficit is worst, and **the selectivity is the bloom’s** — no hemispheric constant was invented to produce it. DJF is austral summer and carries the shortwave: incoming solar there is 460 W m⁻² against 87 in JJA.

Why it is shelved. The tropical cost is 0.0129 W m⁻² per unit S , so the ± 0.5 tolerance binds at $S \approx 39$ — below Woodhouse’s published floor of 43. And at 44 years in combination the seasonal selectivity does not survive: the tropics move almost as much as the Southern Ocean. **No coefficient in the published range works as a tuning knob.** The code and the input stay in the model, inert.

One coupled result is worth keeping: 11F, cancelled at 20 of 50 years, is the only coupled run that ever reached the energy goal (+0.017 over its decade) and it supplied the $1.40\times$ amplification factor of §12. Both are decade-offset from the rest of the family and must not be read as production numbers.

11 The coupled series

30-yr window	net TOA	Siberia JJA	Siberia DJF	DJF soil	SO cloud
110 baseline	+0.97	8.94	−33.21	−12.92	82.93
11A +G4+D2b	+0.78	8.59	−33.16	−13.14	83.50
11B +tanh snow (<i>falsified</i>)	+0.87	10.98	−34.55	−32.08	83.82
11D +fitted snow, SWEMIN30	+0.83	9.51	−33.22	−13.44	83.48
11E +SWEMIN15+K1	+0.77	9.66	−32.82	−12.54	83.65
CERES	—	—	—	—	89.49

11E is the best of the family on every land column and the lowest net TOA, and it is the **only** arm with a positive DJF soil delta against the baseline. 11B is the falsified tanh scheme and is kept in the table because -32.08 K of DJF soil is a useful scale: the diagnostic has ample power for damage of that kind, just not for 1 K, which is why the 30-year window matters.

What 11E does not fix: net TOA is still +0.77 against a goal of zero, and the Southern Ocean cloud fraction is 83.65 % against CERES’s 89.49 — 5.8 pp short, essentially unchanged across the whole family. That is what 11G is for.

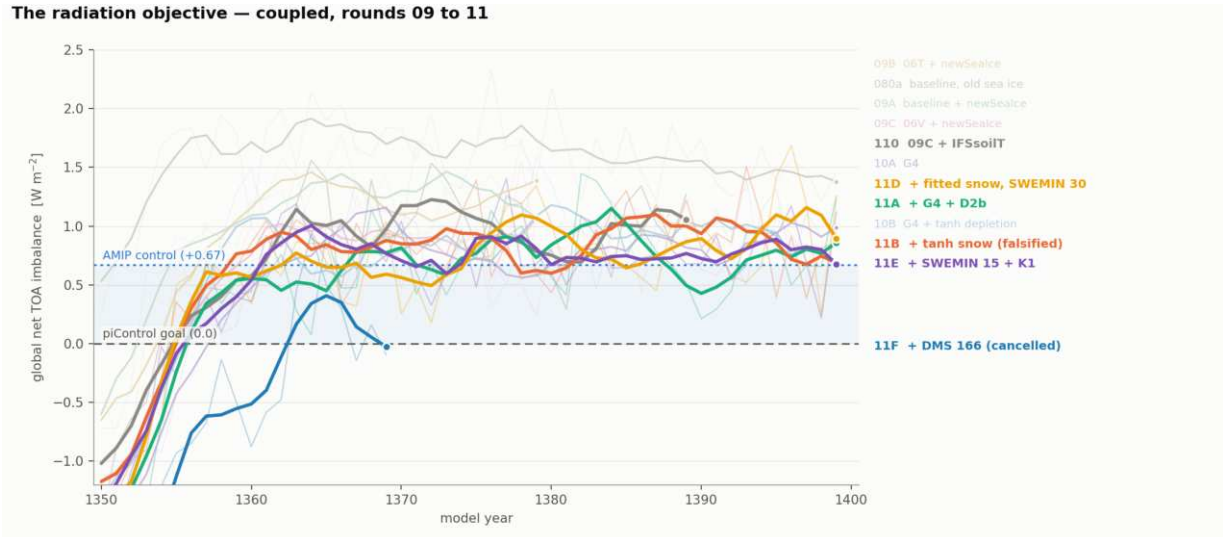


Figure 3: The coupled radiation objective, rounds 09–11. Older generations are drawn thin. The whole 11 family clusters near or above the AMIP control line and none approaches the goal; the first ≈ 10 years of every curve is spin-up from the restart, which is why the 30-year window is used for scoring. 11F (cancelled) is the only curve to reach zero, in a different decade of an unequilibrated spin-up.

12 Coupled amplification: $1.40\times$, and it is sea ice

11F minus 11E over their common 20 years is a clean coupled pair differing only by the DMS coefficient — the same lever measured in AMIP as U1 minus T3. Coupled SO net TOA moved -2.767 against AMIP’s -1.976 : **amplification** $1.40\times$, with SO SST -0.598 K and sea-ice fraction $+3.2$ pp, all resolved.

The mechanism matters more than the factor. The coupled *cloud* response was *smaller* than AMIP’s (-0.44 against -2.25 , not even resolved) because the cloud adjusts away once the surface cools. The amplification comes from **sea-ice albedo, which is clear-sky, not cloud forcing**. So a lever’s AMIP cloud-radiative-effect number is *not* what survives coupling — the surface response is. This is also why S4, which acts on the *surface*, is the right candidate to test coupled, and why 20 years makes 1.40 a lower bound: sea ice adjusts on decadal timescales.

13 Three structural problems

13.1 The AMIP runs are not at 1850 forcing

NCMIPFIXYR=1850 sits in &NAMECECMIP6 — the block 48r1 renamed and no longer reads. The runtime value is -1 and the model reads *yearly* GHG, ozone and aerosol: the log shows 1869 in the first leg and 1911 in the last. **Krakatoa is visible** as a synchronised 1883–86 dip in every AMIP arm (Fig. 4) — a run at fixed 1850 forcing cannot contain a volcano.

Every *lever delta* stands, because both arms of every pair share the forcing. The *absolute* energy baseline does not: mean CO_2 over the window is ≈ 294 ppm against a PI 284.3, worth $\approx +0.15 \text{ W m}^{-2}$, about a quarter of the control's $+0.64$. The coupled runs *are* correctly pinned. A one-line fix in the live namelist was verified to work (run Z1) and is deliberately *not* applied to runs in flight, because it would break comparability with the whole archive; it needs one dedicated control run instead.

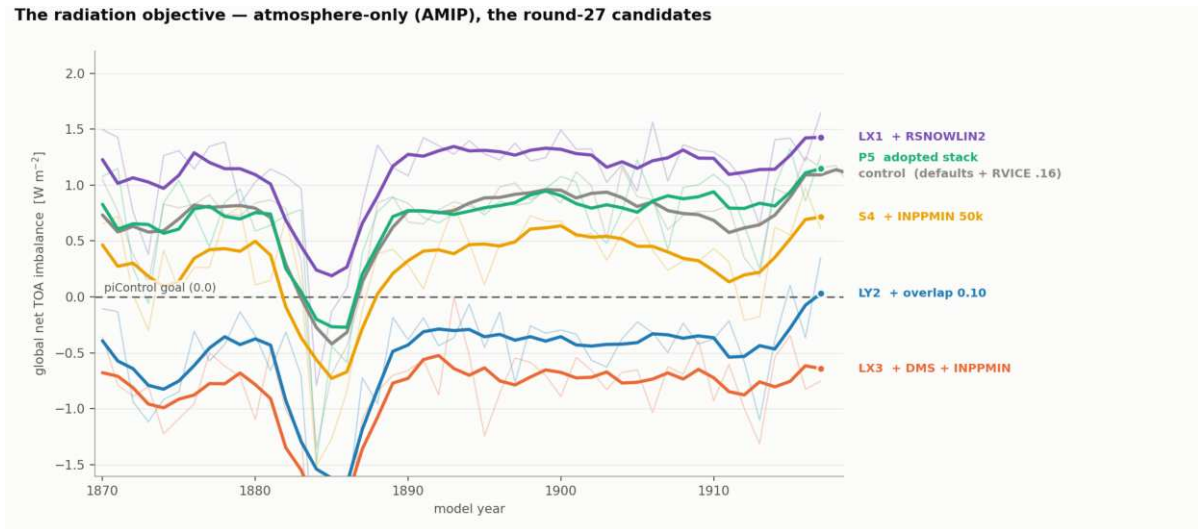


Figure 4: AMIP net TOA by year. The synchronised 1883–86 dip is Krakatoa. S4 lands nearest the pre-industrial goal; LX3 and LY2 overshoot past it; RSNOWLIN2 alone (LX1) moves it the wrong way.

13.2 TOA and surface disagree

Covered in §5: the surface shortwave error exceeds the TOA error everywhere, and no cloud-reflectivity lever can satisfy both. This is the most likely reason the Southern Ocean has resisted 58 runs, and it is one pass over existing data to test — atmospheric absorption is TOA net SW minus surface net SW, model against CERES.

13.3 The error field is dominated by things no lever touches

Global RMSE barely moves between arms (TOA $11.48 \rightarrow 12.40$, surface $14.44 \rightarrow 15.18$) because the pattern is set by the tropical Pacific dipole, the stratocumulus decks off Peru, Namibia and California, and the land signals — all at $\pm 20 \text{ W m}^{-2}$ and all unchanged by everything tried. Against those, **the Southern Ocean box is comparatively well-behaved**.

14 What one year can and cannot tell you

This campaign has retracted 38 claims; roughly 45 % were answerable at a desk in seconds. The thresholds below are the single most useful artefact it has produced, and every number in this summary is quoted against one.

metric	1-year pair	44-year pair
tropical LW CRE	± 0.279	± 0.042
tropical SW CRE	± 0.990	± 0.149
tropical net TOA	± 1.656	± 0.250
SO SW CRE	± 1.974	± 0.298
SO SW CRE, DJF	± 3.82	—
SO LW CRE	± 0.584	± 0.088
global net TOA	± 1.333	± 0.201
Siberia T2m DJF / MAM / JJA / SON	—	± 0.604 / ± 0.404 / ± 0.246 / ± 0.450
Siberia DJF soil, coupled 10 yr / 30 yr		± 0.75 / ± 0.37

Screening at one year works above $\approx 2 \text{ W m}^{-2}$ and inverts below it. A1a reproduces its 44-year Southern Ocean response at one year to within 2%; B3 comes out with the *opposite sign*. It is legitimate for the radiative metrics only — never for temperature, whose seasonal thresholds are 44-year numbers. Two of the four arms in the DMS bracket fell below their thresholds and carried no information at all; reading a trend through them produced a confident wrong answer that had to be withdrawn the same day.

Coupled comparisons need ≥ 30 years. At 10 years the 11E–11D Siberian DJF difference came out with the wrong *sign*.

Check the prior record before designing a round. `preflight.py` reports the source default, namelist reachability, every run and every runscript that has ever set a parameter and to what value, git provenance and prior report mentions. It was scanning one runtime tree and one runscript directory and so said “never set in any run” about two parameters with real AWI history on the same day — `RPRCON`, tuned at TCO319 in 2022, and `RSNOWLIN2`, set to 0.04 in the DARS2 runscripts. It now scans every tree. Its remaining blind spot is the GitHub issues, where the *why* lives and which issue search does not index.

15 Where the energy error actually sits

The global imbalance is small because large regional errors cancel. Period-clean, per band, model minus CERES:

band	net TOA error	channel
Southern Ocean 45–65S	+7.46	shortwave cloud — too little reflection
tropics 30S–30N	−0.67	longwave cloud — too little trapping (−2.16 in LW CRE)
60–90N	+0.24	near-neutral after K1
global	+0.64	the residual of the above

The two regional failures act through different channels, which is why one lever family cannot fix both and why the campaign kept finding that closing one opened the other. It is also why the tropical longwave deficit was promoted to first priority in round 26 — it appeared to *gate* the Southern Ocean fix, since the tropics could not afford the shortwave cost of an SO lever. Round 28 dissolved that framing: the levers are not regionally selective, so the gate was never the binding constraint. What binds is selectivity itself.

15.1 The tropics are too dark, not too bright

Worth stating explicitly because it has caused sign errors, including in this campaign’s own scoring: the tropics sit *below* CERES. They need *more* absorbed energy. A lever that adds tropical reflection is spending an objective, not improving one; a lever that adds tropical absorption is helping. The ± 0.5 tolerance is therefore **one-sided**.

16 Closed by measurement

Kept short; each of these is a hypothesis that a measurement killed, and each is in `report.tex` with the number that killed it.

claim	what killed it
the boreal failure is an LPJG <i>parameter</i> error	forcing transfer reproduced it with identical parameters
larch is excluded by a climate gate	no cold gate; clears GDD5 on 84 % and <i>loses</i> to grass 5.8:1
the extratropical tropopause is displaced	same level in all five bands; it is a sharper transition
cold and dry are independent errors	one entry point; the cold point explains 61 % of the drying
marine CCN can be tuned into the SO fix	no coefficient in the published range clears the tropics
RCL_INPPMIN makes DMS redundant	annual means coincide, seasons do not (DJF -3.72 vs -7.25)
tropical LW cannot be moved	RSNOWLIN2 moves it $+2.74$, 127 % of the deficit
warming cannot restore the forest	offline $+2$ K recovers 59 % of the gap
aerosol is the Southern Ocean's missing reflection	measured OUT: -0.116 , six times smaller than the AOD inference
CERES and ERA5 can both arbitrate SO cloud area	ERA5 <i>is</i> IFS; it shares the model's deficit (-6.31 vs -7.45)
the SO cloud-amount deficit is reachable from CCN	the lifetime route divides by zero at NAERCLD= 0
RKOOPTAU is a cirrus-selective lever	its only use in <code>cloudsc.F90</code> is commented out — a no-op
EC-Earth3's ENTRORG transfers to 48r1	10× formulation difference; their DETRPEN/RPRCON do

17 The path forward

Cheapest-decisive first.

1. 11G — S4 coupled (submitted, job 26868368). 11E with one namelist line changed, a strict one-variable pair against a completed 50-year run. Tests the arm that now looks best on both headline targets, and the $1.40\times$ sea-ice amplification should make its coupled effect *larger* than the AMIP number. Score on a ≥ 30 -year window.

2. Atmospheric shortwave absorption. One pass over existing data. If the model absorbs the wrong amount of shortwave *within* the atmosphere, then the Southern Ocean cannot be fixed by cloud reflectivity at all, and every remaining cloud lever is chasing the wrong error. This is the highest-information item on the list and costs no compute.

3. One fixed-forcing control run to re-baseline the absolute energy target. Do *not* mix fixed- and transient-forcing arms in one comparison; the archive's deltas transfer, only the baseline needs re-establishing.

4. The final land round. It now has what it never had: the $+1/ + 2$ K transfer function, three unread competition experiments, and the knowledge that 50 coupled years shows a transient rather than an equilibrium, so vegetation must be judged offline or over centuries.

Do not spend more runs on: the snow scheme (exhausted), marine CCN as a tuning knob (no published coefficient works), RCL_OVERLAPLIQICE (damages the boreal at 0.10, negligible at 0.35), or single-lever Southern Ocean cloud hunts until item 2 is answered.

18 Assets

For whoever picks this up. Everything is in `investigation_awiesm3_high_lat_cold_bias_global_toa_positive`.

what	where
the full record, 110 pages	<code>report/report.tex</code>
run list — the single source of truth	<code>scripts/analysis/runs.py</code>
the standard evaluation, one command	<code>scripts/analysis/evaluate.sh</code>
has this been tried?	<code>scripts/analysis/preflight.py</code> PARAM
the three-constraint ranking	<code>scripts/analysis/so_decision_table.py</code>
SO and global bias maps	<code>scripts/analysis/{so,global}_pattern_map*.py</code>
net-TOA history, coupled and AMIP	<code>scripts/analysis/plot_net_toa_campaign.py</code>
coupled 11-series scoring	<code>scripts/analysis/coupled_r11_eval.py</code>
coupled vegetation and GDD5	<code>scripts/analysis/coupled_pft_check.py</code>
coupled amplification factor	<code>scripts/analysis/coupled_so_amplification.py</code>
working practice for this model	skill <code>awiesm3</code> ; evaluation protocol <code>amip-eval</code>

Two operational traps that cost real time. `esm_runscripts` prompts interactively if the experiment directory exists, so a non-interactive submit dies with a `prompt_toolkit EOFError` having staged several GB and queued *nothing* — always check `squeue`, and after any resubmission confirm the value you changed actually reached `<run>/work/fort.4`, because a cancelled attempt's `fort.4` survives. And LPJ-GUESS writes *all years per gridcell* before moving to the next, so `tail`-ing a 13 GB `fpc.out` samples the last few *gridcells*, not the last year; stream with `awk` filtered on year and box (13 GB in 20 s).

19 What would change the picture

Stated in advance, so the next round can be scored against them rather than narrated.

- **11G improves coupled net TOA and cools SO SST.** Then S4 is adopted for the spin-up and the Southern Ocean objective is as closed as this campaign can make it.
- **11G cools the SO but does not improve net TOA.** Then the sea-ice feedback is paying for the Southern Ocean out of the global budget, and the two objectives are coupled after all — through the ice, not through the cloud.
- **Atmospheric shortwave absorption turns out to be biased.** Then no cloud reflectivity lever can satisfy TOA and surface together, the Southern Ocean’s resistance to 58 runs is explained, and the target itself needs restating. This is the single most informative thing left and it costs no compute.
- **The offline competition experiments move larch without warming.** Then the forest can be recovered independently of the cold bias, and the land round is a vegetation problem rather than an atmospheric one.
- **The fixed-forcing control lands near +0.5 rather than +0.64.** Then a quarter of the “remaining” imbalance was never model error, and S4’s +0.25 is closer to +0.1.

20 Carried forward

lever	parameters	what it buys
G4	ECE_TUNE_RVRSMIN 1000/1000/225	Siberian JJA, the campaign’s best land lever
K1	12 RVVEGALB + 4 SOILALB= 0.95	snow-free land albedo; tundra excluded
P3	ECE_SNOW_SCF= 3 + 8 parameters	depletion fitted to 36 492 snow courses
P5	ECE_SNOW_SCF_SWEMIN= 15	the autumn mass floor; validated coupled by 11E
D2b	RCL_INPSEA= 0.2	marine ice nuclei
S4	RCL_INPPMIN= 50000	new: SO surface and global energy
<i>Not carried:</i> RSNOWLIN2, ECE_DMS_CCN_SENS, RCL_OVERLAPLIQICE.		

All namelist-only. The DMS ICMCL field and the CCN code path remain in the model and are inert at ECE_DMS_CCN_SENS= 0 — verified bit-for-bit.